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Muscle activation pattern and gait biomechanics after total knee replacement

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Abstract

Objective. The aim of this study is to evaluate residual muscle function abnormalities after total knee replacement, with respect to gait kinematics and kinetics.

Design. Longitudinal study on a follow-up of up to two years.

Background. Gait usually presents an excellent improvement after total knee replacement. Nevertheless, some kinematics and kinetics abnormalities persist even after a long period of time and they might have implications in long-term prosthesis failure. Additionally, lower limb muscle activity has not been sufficiently studied in the past directly by means of dynamic EMG.

Methods. Nine patients who had a posterior cruciate sparing total knee replacement design were evaluated by means of clinical assessment and gait analysis at the end of rehabilitation trials at six, twelve and twenty four months. EMG from trunk and lower limb muscles was registered and elaborated through a statistical detector for the on–off timing.

Results. Gait analysis showed a slow gait, with a "stiff knee gait pattern" and prolonged muscular co-contractions during stance.

Conclusions. Knee kinematics and kinetics abnormalities during loading acceptance after total knee replacement are associated with co-contractions in muscular activation pattern.

Relevance

Gait pattern at two years after total knee replacement is deemed unphysiological, potentially dangerous for the implant duration in time and responsible for residual disability in patients. Muscular behavior during daily activities provides insight into the integration of the prosthetic biomechanics within the muscular-skeletal system. This information is of relevance to improve prosthetic design, rehabilitation programs and knee motor performance.
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Keywords: Total knee prosthesis; Gait analysis; Dynamic EMG; On–off detection; Rehabilitation

1. Introduction

Even in the most clinically successful cases many patients treated by total knee replacement (TKR) cannot achieve normal joint function over time. In most cases gait remains slower than normal, and the treated knee has limited flexion both during the stance and the swing phase. Different patterns of external flexion-extension joint moments have been described. The abnormal characteristics of flexion-extension moments found during gait were thought to be associated to abnormal phasing of quadriceps and hamstrings (Andriacchi, 1993). Dorr et al. (1988) found a continued "stiff knee" during stance associated with increased knee flexion moment, and greater request for quadriceps and biceps activity in the cruciate-sacrificed TKR with respect to the posterior cruciate-retained knee. This gait pattern was attributed to avoidance of shear or to a habit developed before surgery. Wilson et al. (1996) found a decreased knee range of motion in posterior-stabilized prostheses, and two groups of flexion and extension knee moment, with increased quadriceps and hamstrings muscle activity, but no isokinetic strength deficits. Nevertheless, altered muscle function was not considered responsible for decreased knee range of motion during gait, in the presence of functional passive

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range of motion. Andriacchi (1993) found a gait pattern which tended to extend the knee throughout the stance phase thus avoiding the quadriceps demand. The extension pattern at the knee, very similar to that of patients with anterior cruciate ligament injuries, was called "quadriceps avoidance gait" and attributed to proprioception impairment and the disruption of the mechanical advantage mechanism during knee flexion with consequent functional adaptation as a response to factors such as instability or weakness (Andriacchi et al., 1999).

The purpose of this study was to serially follow, by means of gait analysis, up to two years post-operatively, a group of patients treated by total knee replacement and a rehabilitation program aimed at restoring strength in knee muscles, proprioception and gait-reappraisal, to highlight any residual muscular EMG function abnormalities and their relationship with knee biomechanics. In fact, muscle function during gait has not been studied sufficiently in the past directly by means of dynamic electromyography (EMG) (Dorr et al., 1988; Wilson et al., 1996; Benedetti et al., 1999). Findings of the present study can increase the knowledge of the muscular control of knee prosthesis during gait, providing insight into the integration of the prosthetic biomechanics within the musculo-skeletal system. This information is particularly relevant when combined with information about the implant kinematics which could affect knee strength by quadriceps moment arm and antagonist co-contraction (Banks, 2003). Abnormal knee prosthetic biomechanics might have implications in long-term failure of prostheses (Banks et al., 1997; Hilding et al., 1999).

2. Methods

2.1. Subjects

Fifteen patients referring to the Department of Physical Therapy and Rehabilitation were included in the study. Inclusion criteria were no severe arthritic involvement of other joints, and no systemic pathologies. Indication for TKR was primary degenerative osteoarthritis in all the cases. Six patients did not complete the study for concurrent diseases aroused during the follow-up. Nine patients, three males and six females, age ranged 55–74 years (mean age 66.1 (SD 5.2), were evaluated at a follow-up of six, twelve and twenty four months after TKR. Their mean height was cm 159.3 (SD 5.8) and mean weight was kg 70.3 (SD 8.1). The arthroplasty model used was the Interax (by Stryker/Howmedica, Ireland), which is a cementless total condylar implant with posterior cruciate sparing, allowing an early joint motion and weight bearing. The patella was not replaced. Patients were treated by the same surgeon.

2.2. Rehabilitation protocol

Postoperative rehabilitation started the day after the operation. Continuous passive motion was carried out, within a range in which the pain was tolerable, to gradually gain a range of motion (RoM) from 0° to 90°. Muscular strengthening for quadriceps and hamstrings was achieved by means of electrical stimulation, isometric contractions, and active exercises for hip and knee motion. Rehabilitation sessions were carried out twice a day and patients were instructed to continue exercises when they were alone. The patients could bear full body weight on the affected leg after two weeks. During this period, exercises for restoring walking ability were performed using crutches to reduce weight-bearing on the affected leg. The patients were instructed to extend the knee completely at the heel strike and to maintain a physiological rocker on the heel, ankle and forefoot during the stance phase of walking. A simplified proprioceptive re-education program was followed, first through open kinetic chain exercises, then through closed kinetic chain exercises in a sitting position, and finally while standing. When patients were discharged from the hospital, they were told to perform daily exercises at home. They also received counseling for proper walking, stair climbing, and weight lifting and avoiding excessive knee stresses. We examined patients at six, twelve, and twenty-four months after surgery. At each follow-up stage they underwent a two week period of rehabilitation at the hospital, keeping up exercises to maintain the knee range of motion, to strengthen quadriceps and hamstrings and to optimize walking ability. Gait analysis was performed at the end of each rehabilitation period.

2.3. Clinical assessment

A modified rating scale (IKS) suggested by Insall et al. (1989), consisting of knee clinical and functional scores, was adopted. The knee clinical score is based on the assessment of active RoM, pain, knee stability, flexion contractures, and knee alignment, while the functional score is based on support use, walking and stair climbing ability. The range for clinical score and functional score is 0–100 (a score between 85 and 100 is considered excellent). Muscle strength was assessed using a 0–5 scoring system for manual muscle testing (Kendall and Kendall McCreary, 1983).

2.4. Gait analysis methodology

Lower limb functional evaluation during gait was performed using the ELITE stereophotogrammetric system (BTS, Milano, Italy) and a Kistler platform (Kistler Instrument, Winterthur, Switzerland) for the acquisition of kinematic and kinetic variables. The gait of the patients was assessed on the treated limb in three consec-

utive trials at free speed and with comfortable sport shoes. Each patient worn the same shoes throughout gait analysis follow-up and the rehabilitation sessions. The CAST (Calibrated Anatomical System Technique) experimental protocol was used for gait data elaboration following the reference system proposed by Cappozzo et al. (1995).

Kinematic and kinetic data were then reduced by means of a parametric analysis, and compared with those obtained from a control population of ten healthy subjects (six males and four females) (Benedetti et al., 1998), age ranged from 55 to 72 years (mean age 66.9 (SD 5.8)), mean height cm 163.7 (SD 8.1) and mean weight kg 66.2 (SD 9.6). The repeated measure ANOVA was used ($P < 0.01$) to compare the three groups, if the Levene test ($P < 0.05$) for uniformity of variances was not significant. Otherwise the Friedman Test non-parametric test ($P < 0.05$) was used. SPSS was used for all statistical analysis methods. The surface myoelectric signal was recorded by the TELEMG electromyograph (BTS). Eight muscles were simultaneously examined: the ipsilateral and contralateral erector spinae (IES, CES), the gluteus medius (GM), the rectus femoris (RF), the medial (MHAM) and lateral (LHAM) hamstrings, the gastrocnemius (medial head) (GAS), and the tibialis anterior (TA). The myoelectric signals were acquired simultaneously with the kinematic and kinetic data, then processed off-line by means of a statistical detection algorithm to obtain muscle on–off timing (Bonato et al., 1998; Benedetti et al., 1999).

3. Results

3.1. Clinical assessment

No statistical difference was found in subject's height and weight. The clinical and functional scores were excellent throughout the follow-up (Table 1). Notwithstanding, after 24 months a significant decrease in the functional score was reported ($P < 0.004$). The items that influenced the lowering of the score were evaluated: in seven patients there was a decreased ability to walk unlimited and/or descending stairs without a rail. No significant difference in the measurement of the knee flexion at different follow-ups was found. In five patients a slight pain in the contralateral knee was reported. Manual muscle testing showed a significant improvement in the quadriceps ($P = 0.005$) and tibialis anterior ($P = 0.01$) strength at 12 months post-op. Triceps strength remained reduced throughout the follow-up.

3.2. Gait analysis

3.2.1. Time–distance parameters

Data relative to time and distance parameters during the three follow-up exams, compared to the control are reported in Table 2. The stance phase was significantly increased at all the follow-up exams ($P = 0.03$). Stride length, cadence and, as a consequence, speed of progression were significantly reduced throughout the follow-up ($P < 0.0001$).

3.2.2. Knee kinematics and kinetics

Knee flexion during midstance (K2) was significantly reduced throughout the follow-up ($P = 0.001$) (Table 3), so as the flexion at toe-off (K4) ($P = 0.04$) and the flexion during swing phase (K5) ($P < 0.0001$). As a consequence the total range of motion was reduced ($P < 0.0001$). A significant decreasing in both the first and the second external extension moment peaks (KM1, KM3) was found ($P < 0.0001$) (Table 3). The flexion moment (KM2) was not significantly altered. On the coronal plane the knee abduction moment (KM4) resulted increased ($P = 0.005$) throughout the follow-up while adduction moment (KM5, KM6) were decreased ($P < 0.0001$, $P = 0.001$).

3.2.3. Dynamic EMG

The electromyographic study (Fig. 1) showed an increased use of the ipsi and contralateral erector spinae, both during the stance and the swing phase, suggesting

Table 1
Clinical data on TKR

Parameters	6 Months		12 Months		24 Months		ANOVA *Friedman Test
	Mean	SD	Mean	SD	Mean	SD	*P*
Clinicall functional score							
Clinical score*	92	5.7	95	3.2	93	6.3	NS
Functional score*	89	10.1	92	10.7	85	13.9	0.004
Knee flexion (deg)	104	12	105	12.7	104	11	NS
Manual muscle test for strength							
Quadriceps*	4.2	0.4	4.7	0.5	4.5	0.5	0.005
Tibialis anterior*	4.4	0.5	4.6	0.5	4.8	0.4	0.01
Hamstrings	4.4	0.5	4.5	0.5	4.5	0.5	NS
Triceps	4.2	0.6	4.3	0.5	4	0.6	NS

Table 2
Time distance parameters

Parameters	6 Months		12 Months		24 Months		Control		ANOVA *Friedman Test
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	*P*
Stance (%GC)*	62.4	3	62.3	2.1	62.6	2.2	60.8	1.2	0.003
Stride length (cm)*	106.7	15	112.2	14	112	15.3	130.9	8.5	<0.0001
Stride length (%height)*	64.5	9.6	70.8	7.7	70.3	7.9	80.5	5.4	<0.0001
Gait cycle duration (ms)	1.3	0.1	1.2	0.1	1.2	0.1	1.08	0.1	<0.0001
Cadence (str/min)	47.6	3.9	51.6	3.9	51.2	3.3	55.9	4.7	<0.0001
Speed of progression (cm/s)	85.1	15.9	96.7	15.7	95.8	16.4	122.1	13.7	<0.0001

Table 3
Knee kinematics and kinetics

Parameters	6 Months		12 Months		24 Months		Control		ANOVA *Friedman Test
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	*P*
Degrees									
K1 Flexion heel strike	3.4	5.9	2.3	3.5	4.2	5.4	1.5	5.6	NS
K2 Max flexion loading response	10.4	7.1	12.1	5.1	11.5	6.4	16.7	5.6	0.001
K3 Max extension stance	4.2	6	5.9	4.7	5.5	6.6	6.1	5	NS
K4 Flexion toe off	33.7	7.5	36.1	6.7	33.9	6.1	38.2	5.3	0.04
K5 Max flexion swing	52.9	8.2	55.7	7.8	54.3	8.1	63.3	4.3	<0.0001
K6 Total sagittal plane excursion	48.9	5.2	49.7	5.3	48.8	7.1	57.1	5.4	<0.0001
*%BW*H*									
KM1 1st max extension moment	−2.3	1.1	−2.5	1.1	−2.6	1.1	−4.7	1.6	<0.0001
KM2 max flexion moment	0.7	1.2	0.9	1.4	0.5	0.8	0.7	1.3	NS
KM3 2nd max extension moment	−1.3	1.2	−1.3	0.9	−1.3	1	−2.6	1.1	<0.0001
KM4 max abduction moment	1.2	0.7	1	0.7	1.5	0.9	0.9	0.6	=0.005
KM5 1st max adduction moment	−0.9	0.7	−2	1.6	−1.2	1.3	−2.7	1.2	<0.0001
KM6 2nd max adduction moment	−0.6	1.2	−1.5	1.9	−0.7	1.7	−2.0	1	=0.001

compensatory mechanisms in the trunk. There is no evident trend to reduce this abnormal activity during the follow up. In most patients a prolonged activity of rectus femoris, hamstrings and tibialis anterior during stance was reported throughout the follow-up. Furthermore a premature activation of the gastrocnemius after 6 months from surgery was present. This pattern became closer to control during the following exams.

4. Discussion

Despite the rehabilitation training followed by all the patients to improve the quadriceps strength, proprioception and gait performance, 6–24 months after surgery, gait abnormalities were present in patients with excellent clinical score. Moreover a significant decreasing of functional score is present at 24 months after surgery. This result can be explained by the reluctance to perform daily rehabilitation exercises at home in the second year after surgery as most patients reported. In our study, TKR gait pattern was characterized by slow speed of progression with reduced stride length and reduced knee flexion during load absorption phase, propulsion and swing phase, associated with reduced external extension moment and prolonged co-contraction of rectus femoris-hamstrings and gastro-tibialis anterior. Moreover the abnormal muscular activity at the trunk ipsilateral to the replaced knee and the prolonged tibialis anterior activity during mid stance could support the hypothesis of a compensative mechanism on the coronal plane aimed at reducing the adduction moment at the knee, as documented by kinetics. Since the kinematic and kinetic gait alterations highlighted in this study were consistent with data from the literature for different models of TKR (Andriacchi et al., 1982; Dorr et al., 1988; Wilson et al., 1996), they seem to be independent from the peculiarities of the geometrical design. In particular the posterior cruciate sparing design evaluated in this study does not avoid the greater muscular activity request.

Prolonged activity in knee muscles found by means of dynamic EMG, and not emphasized in previous studies, seems to be a peculiar feature of TKR gait, persistent throughout follow-up, associated with the "stiff-knee pattern". Since pain, severe knee flexion contracture or severe quadriceps weakness were not present in the patients, muscular co-contractions around the knee seem to account for a compensatory mechanism aimed

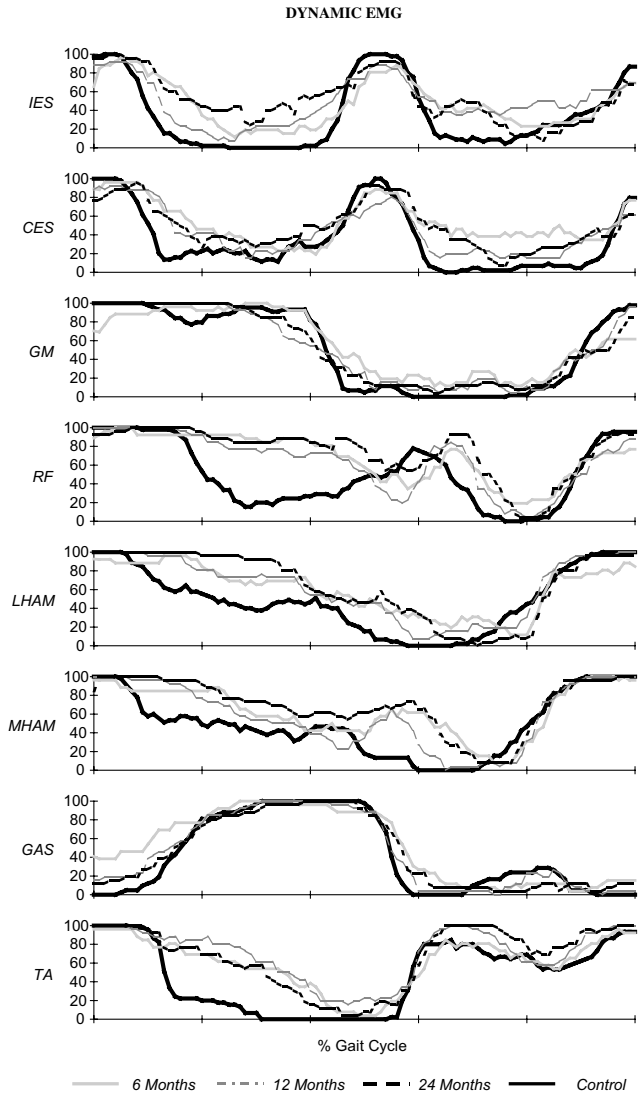

Fig. 1. On–off muscle timing percentage of gait cycle. The pattern of muscular activity is presented as percentage of number of patients for each muscle at different stages of the follow-up compared to control data. IES, ipsilateral erector spinae; CES, contralateral erector spinae; GM, gluteus medius; RF, rectus femoris; MHAM, medial hamstrings; LHAM, lateral hamstrings; GAS, gastrocnemius; TA, tibialis anterior.

at providing greater control of knee kinematics during stance phase (McGuinty et al., 2000; Banks, 2003).

Recent studies based on fluoroscopic analysis demonstrated unpredictable intrinsic knee kinematics both in fixed and mobile insert implants (Banks et al., 1997; Stiehl et al., 2001; Dennis et al., 2003). In many patients the presence of paradoxical pattern of anterior femur translation during knee flexion was reported. These results suggest that rarely physiologic knee kinematics can be obtained in vivo without the intact control mechanism of the anterior and posterior cruciate ligaments. The presence of muscle co-contraction around a "stiff knee" during stance might be the resultant of this abnormal prosthetic biomechanics in terms of functional adaptation of the patient to a sense of instability.

Moreover, many authors have underlined the role of the proprioceptive deficit that may have developed both before and during surgery (Cash et al., 1996; Barret et al., 1991; Sharma and Pai, 1997) although proprioception is not easy to measure. The decline of proprioception with aging has been claimed responsible for arthritic knee changes through an altered muscular coordination pattern in the control of joint loading. In this hypothesis muscle function abnormalities should be considered an effort to maximize proprioceptive input.

Finally, another hypothesis is that the "stiff knee pattern" is a result of a habit related to pain and functional impairment developed before surgery. Fisher et al. (1997) found that the gait pattern of knee arthritic patients, very similar to that of TKR and anterior cruciate ligament patients, remained unchanged after a rehabilitation protocol aimed at restoring muscular strength, endurance and contraction speed and concluded that the patients would have developed a strategy of "functional adaptation", acquired over time and became habitual, aimed at reducing pain in the knee, and unresponsive to the rehabilitation protocol.

Results from this study does not allow to demonstrate whether the presence of muscular co-contractions throughout the considered follow-up is influenced from the rehabilitation protocol proposed since a comparison group not receiving rehabilitation was not studied.

In conclusion, (1) the study of surface dynamic EMG elaborated with a statistically reliable detector for the analysis of the on–off muscle timing, has allowed to enlighten muscular activation irregularities during walking in TKR patients. (2) Prolonged muscular co-contraction are associated to "stiff knee pattern" during stance. Future studies on dynamic EMG combined with in vivo kinematic studies by means of fluoroscopy could provide further insight into the integration of the prosthetic biomechanics with the muscular-skeletal system and the relationship between knee prosthesis "intra-articular" kinematics and muscular co-contractions (Fantozzi et al., 2003).

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